## INTRODUCTION

One of the biggest challenges faced by many small to large scale industries are their employee attrition. Understanding those factors that cause attrition depend and vary upon various factors that comprises it's demographic, social, professional factors.

This study tries to analyse and interpret IBM HR dataset to explore particularly how common professional factors significantly influences employee attrition in the company in context of IBM using descriptive statistical analysis, hypothesis testing and visualisations too.

The findings will provide insights how monthly income, job satisfaction and work life balance, three professional pillars for employee retention in a company affects attrition in IBM.

The ultimate goal of this report is to provide a comprehensive analysis to suggest HR and help in decision making processes in this sector to shape enhanced employee retention strategies.

## HYPOTHESIS

This analysis report focuses on how professional factors act as a push or pull factors for employee attrition.

*Monthly Income:* Acts as a pull factor in general if employee have better opportunities to choose that might affect attrition.

*Work-Life Balance:* Accordingly, monthly income can also affect the push factor, the way you balance your work-life by doing overwork, seeking other job or dealing with daily life issues that might be directly or indirectly related to your economic domain.

*Job Satisfaction:* This all can affect the overall job satisfaction that might affect the attrition employee acting as another push factor.

HYPOTHESIS 1:

**Null Hypothesis ($N_o$):** There is no significant difference between the monthly income between employees who have left the company and who have not.

**Alternate Hypothesis ($N_1$):** There is a significant difference between the monthly income between employees who have left the company and who have not.

HYPOTHESIS 2:

**Null Hypothesis ($N_o$):** There is no association between work-life balance and attrition.

**Alternate Hypothesis ($N_1$):** There is an association between work-life balance and attrition.

HYPOTHESIS 3:

**Null Hypothesis ($N_o$):** There is no association between job satisfaction and attrition.

**Alternate Hypothesis ($N_1$):** There is an association between job satisfaction and attrition.

## STATISTICAL ANALYSIS OF ATTRITION

*General statistical analysis of data and other factors affecting attrition*

Analysing the data it is found that company has a 16.12% of attrition whereas a healthy attrition should be around 10%. 60% of the employees are male and 40% are females. 70.95 and majority of the workers travel rarely. Highest involvement is seen in Research and Development department with 961 employees followed by sales and then HR. 38.91% of workers are bachelor's forming the majority followed by 27.07% masters. 571 out 1470 either least or less satisfied with their environment. Sales representative are prone to attrition whereas least attrition is seen in the role of research director. 45.78% of employees or majority are married. Highest attrition is seen in Single followed by married then by divorced. Highest attrition is seen in the people with poor work life balance and among age groups of 15-25 and least in the age groups of 35-45.

*Analysis of professional factors influencing attrition*

Hypothesis 1: Attrition and Monthly income

The null hypothesis suggests that there is no significant difference between the monthly income between employees who have left the company and who have not. To check its validity we can proceed with t-test and check for the p-value. If the p-value is greater than the significance/p-value of 0.05 then, we accept the null hypothesis.

**Group Statistics**

| | Attrition | N | Mean | Std. Deviation | Std. Error Mean |
|---|---|---|---|---|---|
| MonthlyIncome | Yes | 237 | 4787.09 | 3640.210 | 236.457 |
| | No | 1233 | 6832.74 | 4818.208 | 137.216 |

**Independent Samples Test- Monthly income and Attrition**

| | | | MonthlyIncome | |
|---|---|---|---|---|
| | | | Equal variances assumed | Equal variances not assumed |
| Levene's Test for Equality of Variances | F | | 26.313 | |
| | Sig. | | .000 | |
| t-test for Equality of Means | t | | -6.204 | -7.483 |
| | df | | 1468 | 412.741 |
| | Sig. (2-tailed) | | .000 | .000 |
| | Mean Difference | | -2045.647 | -2045.647 |
| | Std. Error Difference | | 329.734 | 273.386 |
| | 95% Confidence Interval of the Difference | Lower | -2692.446 | -2583.050 |
| | | Upper | -1398.847 | -1508.244 |

As the significance value is 0.00 it is less than 0.05 thus, our null hypothesis is rejected. So, we accept our alternate hypothesis that there is a significant difference between the monthly income between employees who have left the company and who have not.

## Hypothesis 2: Work-life balance and Attrition

The null hypothesis suggests that there is no association between work-life balance and attrition. To check its validity we can proceed with chi-square test and check for the pearson-value. If the pearson-value is greater than the significance/pearson-value of 0.05 then, we accept the null hypothesis.

**WorkLifeBalance * Attrition Crosstabulation**

Count

| | | Attrition | | |
|---|---|---|---|---|
| | | Yes | No | Total |
| WorkLifeBalance | Bad | 25 | 55 | 80 |
| | Good | 58 | 286 | 344 |
| | Better | 127 | 766 | 893 |
| | Best | 27 | 126 | 153 |
| Total | | 237 | 1233 | 1470 |

**Chi-Square Tests**

| | Value | df | Asymptotic Significance (2-sided) |
|---|---|---|---|
| Pearson Chi-Square | 16.325[a] | 3 | .001 |
| Likelihood Ratio | 14.073 | 3 | .003 |
| Linear-by-Linear Association | 6.006 | 1 | .014 |
| N of Valid Cases | 1470 | | |

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 12.90.

As the significance value is 0.001, it is less than 0.05 thus, our null hypothesis is rejected. So, we accept our alternate hypothesis that there is an association between work-life balance and attrition.

## Hypothesis 3: Job satisfaction and Attrition

The null hypothesis suggests that there is no association between Job satisfaction and attrition. To check its validity we can proceed with chi-square test and check for the pearson-value. If the pearson-value is greater than the significance/pearson-value of 0.05 then, we accept the null hypothesis.

**JobSatisfaction * Attrition Crosstabulation**

Count

| | | Attrition | | |
|---|---|---|---|---|
| | | Yes | No | Total |
| JobSatisfaction | Low | 66 | 223 | 289 |
| | Medium | 46 | 234 | 280 |
| | High | 73 | 369 | 442 |
| | Very High | 52 | 407 | 459 |
| Total | | 237 | 1233 | 1470 |

**Chi-Square Tests**

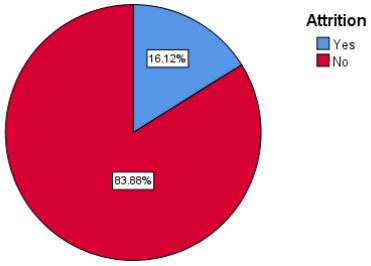
| | Value | df | Asymptotic Significance (2-sided) |
|---|---|---|---|
| Pearson Chi-Square | 17.505[a] | 3 | .001 |
| Likelihood Ratio | 17.356 | 3 | .001 |
| Linear-by-Linear Association | 15.731 | 1 | .000 |
| N of Valid Cases | 1470 | | |

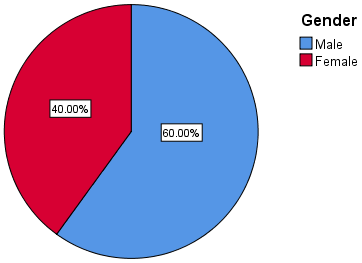
a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 45.14.

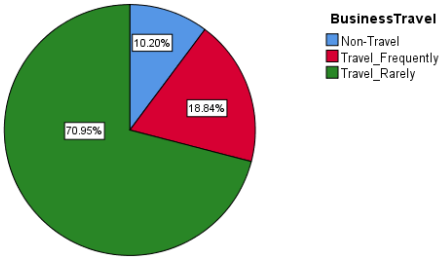
As the significance value is 0.001, it is less than 0.05 thus, our null hypothesis is rejected. So, we accept our alternate hypothesis that there is an association between Job satisfaction and attrition.

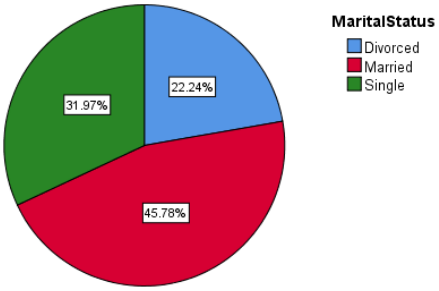
## APPROPRIATE VISUALISATIONS

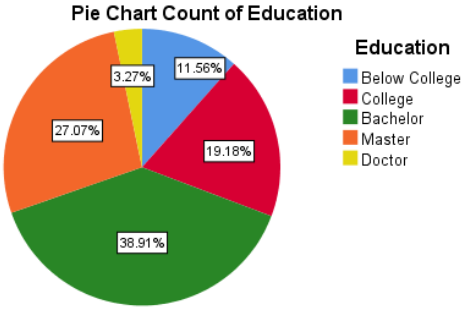
*Visual representation of general factors*

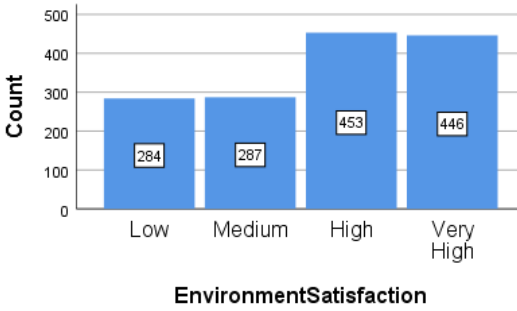


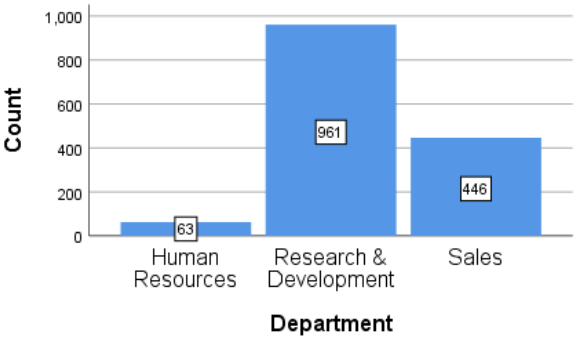



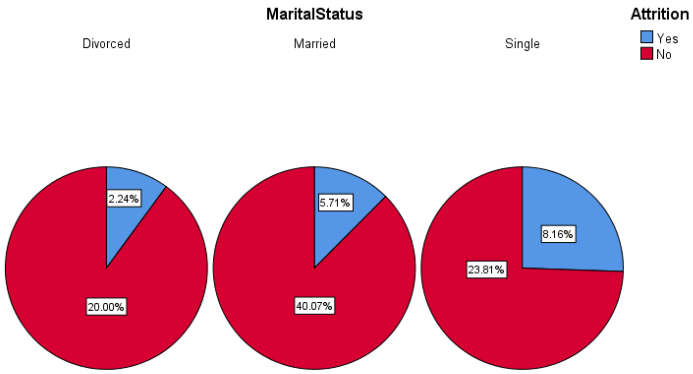



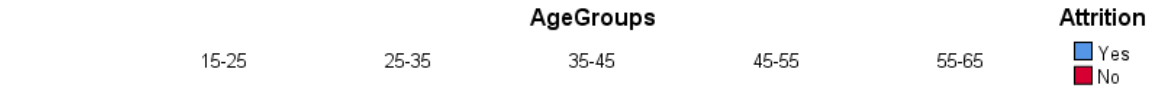



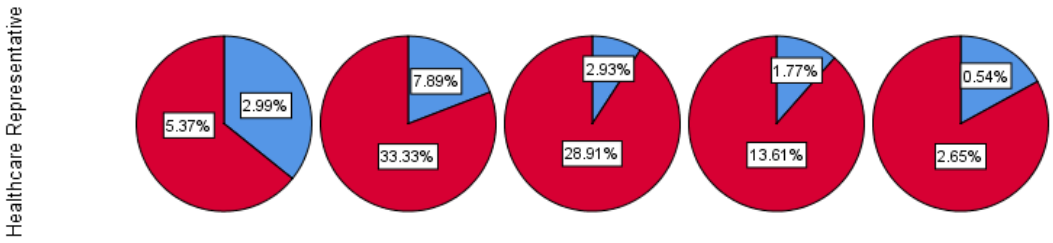



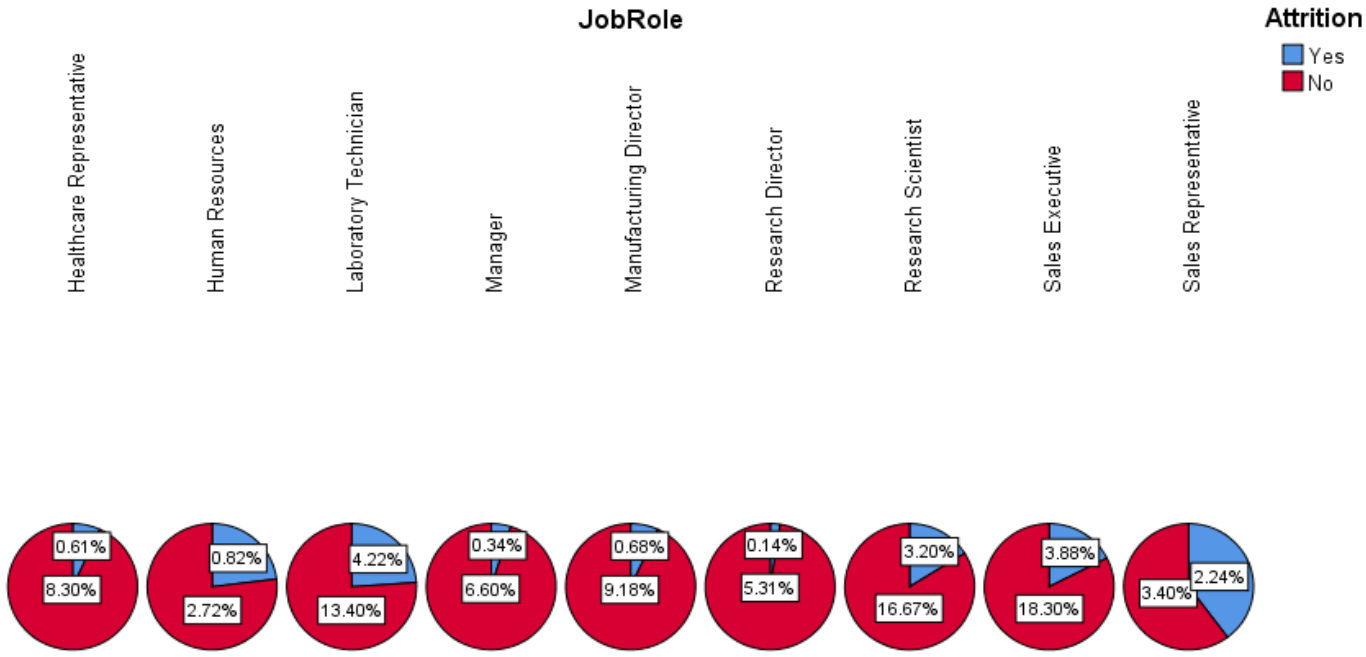



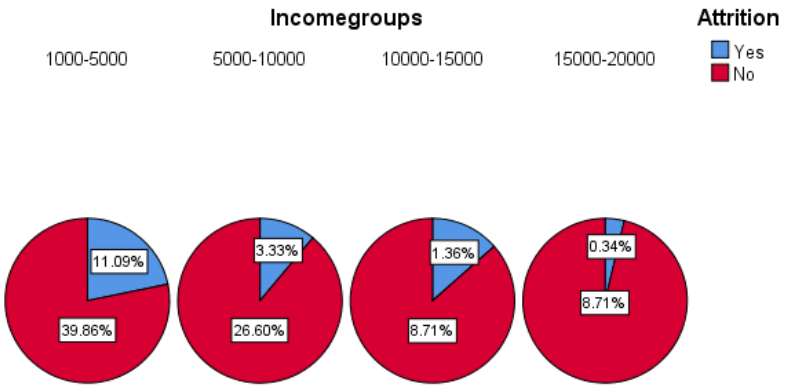



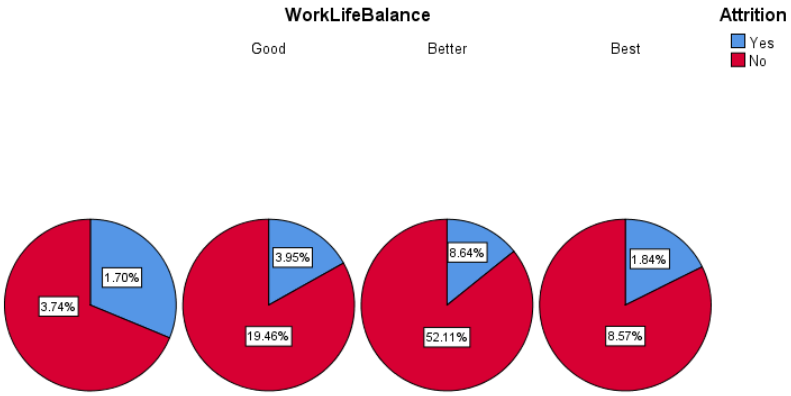





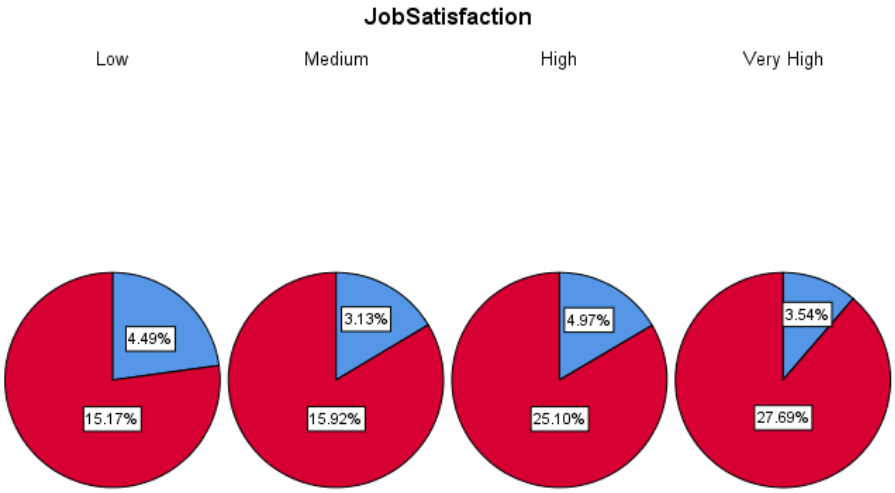

*Visual representation of professional factors*




Lowest-income earners showing highest attrition while highest ones showed the least.




Bad work-life balance showing highest attrition which showed a significant decrease with improvement.




Less satisfied employees showing highest attrition while very high satisfied ones showed the least.

## INSIGHTS AND CONCLUSION

*General Insight*

There are various other factors other than professional domain which are affecting the attrition like age, marital status, job role etc. This can develop general insight that most of the attrition is seen in lower age groups and unmarried because they have much to explore and has risk-taking ability. In context of job roles which might potentially govern the factors like monthly income and satisfaction etc. and highest attrition is seen in sales representative and least in manager. So fragility of roles also drives attrition.

*Professional Domain Insight*

As mentioned earlier, lower age groups, unmarried and vulnerable job roles face greater attrition because these have less job satisfaction and other factors driving that leading to more risk-taking. Better income gives a chance to have adequate resources thus, providing a chance to have a balanced work life due to suffice resources to lead that. Thus, higher probability of higher job satisfaction. Better job roles and income need experience and consistency which these age-groups lack. This financial and social vulnerability leads them to a poor work-life balance and hence constantly finding better opportunities leading to attrition. When there is a poor work-life balance seen it leads them to lesser overall job satisfaction and vulnerable to attrition. Thus, monthly income can lead to a poor work-life balance leading to lesser overall job satisfaction.

***Area to focus:***

- An increment in the admission of experienced employees might help in reducing attrition
- Better career opportunities and pay might act as a pull factor for workers especially for lower-age groups.
- Providing workshops and opportunities to understand the value of work-life balance might increase both productivity and retention of the employees.
- Making the working more interactive and growth seeking might improve the condition of overall job satisfaction

## CONCLUSION

Conclusively, other than professional factors there are various other factors like demography, social or personal that have a role to play in attrition. But talking professionally, monthly income, work-life balance and job satisfaction play a major role in it. Providing more innovative working conditions and methods will improve the physical as well as mental health of the employees leading to lesser attrition, better outputs and profits.

LINK: https://drive.google.com/drive/folders/1i6LBLUB6s3jQ_Y9FHdtJQVYci8LO7_kT